# Self-Resolving Prediction Markets for Unverifiable Outcomes


Vilhjalmur A. Vilhjalmsson (威廉)

Tsinghua University

*weil25@mails.tsinghua.edu.cn*


Algorithm Economics (80470223-0) - Course Project

May 31, 2026

# Presentation Overview

# Introduction

- Name: Self-Resolving Prediction Markets for Unverifiable Outcomes
- Main Topics:
    - Prediction Markets
    - Mechanism Design
    - Information Aggregation
    - Peer Prediction
    - Information Elicitation

**Self-Resolving Prediction Markets for Unverifiable Outcomes**


**Siddarth Srinivasan**
Harvard University
Cambridge, MA, USA
ssrinivasan@seas.harvard.edu

**Ezra Karger**
Federal Reserve Bank of Chicago
Chicago, IL, USA
ezra.karger@chi.frb.org

**Yiling Chen**
Harvard University
Cambridge, MA, USA
yiling@seas.harvard.edu


**Abstract**


Prediction markets elicit and aggregate beliefs by paying agents based on how close their predictions are to a verifiable future outcome. However, outcomes of many important questions are difficult to verify or *unverifiable*, in that the ground truth may be hard or impossible to access. We present a novel incentive-compatible prediction market mechanism to elicit and efficiently aggregate information from a pool of agents *without observing the outcome*, by paying agents the negative cross-entropy between their prediction and that of a carefully chosen reference agent. Our key insight is that a reference agent with access to more information can serve as a reasonable proxy for the ground truth. We use this insight to propose self-resolving prediction markets that terminate with some probability after every report and pay all but a few agents based on the final prediction. The final agent is chosen as the reference agent since they observe the full history of market forecasts, and thus have more information by design. We show that it is a perfect Bayesian equilibrium (PBE) for all agents to report truthfully in our mechanism and to believe that all other agents report truthfully. Although primarily of interest for unverifiable outcomes, this design is also applicable for verifiable outcomes.


Figure: Paper cover.

# Motivation and Problem Setup

- Prediction Markets are excellent at forecasting things that eventually resolve.
  - Pay people once the truth is revealed.
  - E.g., elections, sports, disease spread
- What about questions that don't resolve cleanly?
  - E.g., Content moderation decisions, Causal effects of policies

# Motivation and Problem Setup

## Key Idea

Can we replace the verifiable outcome with a reference agent that has more information about the outcome?

# What is a Prediction Market?

**Definition (What is a Prediction Market)**

A prediction market is a mechanism that allows people to bet on the outcome of an event.

**Example**

- Polymarket
- The stock market
- Sport betting

# Background: Scoring Rules

## Idea

A (proper) scoring rule pays an agent for a probabilistic prediction so that reporting their *true belief* maximizes their expected payoff.

- Problem: this requires observing the outcome $i$.

## Definition (Cross-Entropy Scoring Rule)

Score a report $q$ against a reference prediction $r \in \Delta\Omega_Y$ instead of an outcome:

$$S_{CE}(r, q) = -H(r, q) = \sum_{i=1}^{Y} r_i \log q_i.$$

# Background: The Key Swap

## Standard scoring rule

prediction vs. outcome

$$S(i, q) = \log q_i$$

*Needs the truth to be revealed.*

## Cross-entropy scoring rule

prediction vs. reference

$$S_{CE}(r, q) = \sum_i r_i \log q_i$$

*Needs a reference prediction r.*

This swap is what lets the market resolve *without an outcome*.

# The Self-Resolving Prediction Market Mechanism

- Consider a prediction market for a binary outcome $Y \in \{0, 1\}$ with a common prior $\pi$.
- $M$ agents arrive in sequence.
- Each agent has a private signal $s_i \in [0, 1]$ that is conditionally independent of the others given $Y$.
- Each agent sees the previous market report plus its own signal, forms a belief, and reports it.
- The market terminates with probability $\alpha$ after each report; the terminal agent is the reference.
- Agents $1..T - k$ are paid the cross-entropy market scoring rule against the reference, and the last $k$ agents receive a flat fee $R$.

# The Self-Resolving Prediction Market Mechanism

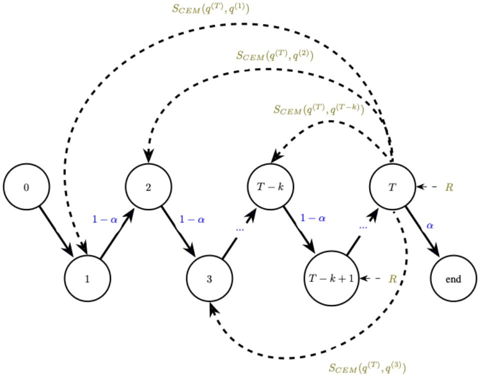



Figure: The Self-Resolving Prediction Market Mechanism.

# Scoring rule in mechanism

- The scoring rule in the mechanism is the CE-MSR.
- The scoring rule is defined as:

$$S_{CEM}(q^t, q^{t-1}, r) = -H(r, q^t) + H(r, q^{t-1}) = \sum_{i=1}^{Y} r_i \log \frac{q_i^t}{q_i^{t-1}}$$

- Intuition: how much better is the new report than the previous one, compared to the reference?

# What the Guarantees Rest On: Five Assumptions

The mechanism's truthfulness and aggregation guarantees are *theoretical*: they rely on five assumptions about agents and their signals.

- **A1 Rationality & risk-neutrality.** Agents are fully rational, risk-neutral Bayesians — and this is common knowledge.
- **A2 Common prior.** Signals $X^{(t)}$ and outcome $Y$ are drawn from a common prior that is common knowledge.
- **A3 Stochastic relevance.** Each signal induces a *unique* posterior, so agents can invert others' reports ("distinct signals").
- **A4 Conditional independence.** Signals are independent given $Y$ — i.e. informational *substitutes*.
- **A5 $(\delta, \eta)$-informative signals.** A floor on how informative each signal is ($\delta$) and a floor on signal probabilities ($\eta$).

Our two experiments probe what happens when these hold — and when they don't.

# What the Paper Proves: Three Guarantees

All three say the same thing from different angles: a well-informed reference makes honesty optimal.

## Thm 1 — A report's influence is bounded

If the reference observes $k$ informational substitutes that agent $t$ cannot, then $t$'s influence on the reference's posterior — the adjustment $\Delta$ — shrinks: $|\Delta| \leq \varepsilon$ once $k \gtrsim \frac{1}{-\log(1-\delta)} \log \frac{1}{\varepsilon}$. *Enough substitutes* $\Rightarrow$ *reference* $\approx$ *ground truth.*

## Thm 2 — Truthfulness becomes strictly optimal

Paid by cross-entropy scoring against a truthful reference, agent $t$ *strictly* maximizes expected payoff by reporting truthfully once the reference has $\geq k$ signals (an explicit threshold in $\delta, \eta, \tau$ and the prior).

## Thm 3 — The gain from lying vanishes

The gain from *any* deviation is bounded by $D_\eta(\Delta, \mathbf{y})$; since $\Delta \to 0$ as $k$ grows (Thm 1), this bound $\to 0$ — and crucially, no $\tau$ is needed.

# Limitations and Proposed Research Questions

The paper's incentive guarantees are *theoretical*: they rest on the reference being a good proxy for the truth, and the mechanism is never tested on real data.

### RQ1 — Simulation (incentives)

Under strategic deviation, *when* is truthful reporting a best response, and how does this depend on how well-informed the reference is?

### RQ2 — Empirical (real markets)

How closely do self-resolving payouts track the standard verifiable (outcome-based) payouts on resolved real-world markets?

# RQ1: Testing the Theory on Its Own Terms

The simulator is built so that A1–A4 hold by construction:

- **A1** agents are Bayesian by construction; the focal agent deviates *within* that rational frame (via $\gamma$).
- **A2** a single common prior $\pi$ generates every signal.
- **A3–A4** signals are distinct, conditionally independent draws given $Y$.

## RQ1 stress-tests A5 (informativeness)

The one assumption we *vary* is how informed the reference is. Sweeping $k$ (the reference's informational substitutes) asks: how informed must the reference be before truthful reporting is the best response? (Thm. 3: deviation gain $\to 0$ as $k$ grows.)

Clean world $\Rightarrow$ isolates the mechanism's core incentive claim.

# RQ1 Preliminaries: The Three Knobs

What the simulation actually varies

## The paper's informativeness parameters (Assumption 5)

- $\delta$ — how informative a signal is. Via the Bhattacharyya coefficient, $\delta = 1 - \mathrm{BC}$: $\delta = 0$ means $P(X \mid Y{=}0) = P(X \mid Y{=}1)$ (useless); larger $\delta$ separates the two classes.
- $\eta$ — a floor on signal probabilities, $\min_{x,\omega} P(X{=}x \mid Y{=}\omega) \geq \eta$. No signal is arbitrarily decisive ($\eta$ bounds how far one report can move a belief).

## Our deviation parameter (simulation construct)

$\gamma$ scales the focal agent's signal: report $\mathrm{logit}(\pi) + \gamma\,\ell$. $\gamma = 1$ truthful, $\gamma > 1$ over-confident.

Link: the theory bounds the *gain* from any deviation ($D_\eta(\Delta, \mathbf{y})$, Thm. 3) and shows it $\to 0$ as $k$ grows. $\gamma$ is how we *realize* a deviation; $\gamma^\star \to 1$ is that bound biting.

# RQ1 (Simulation): Is Truthful Reporting a Best Response?

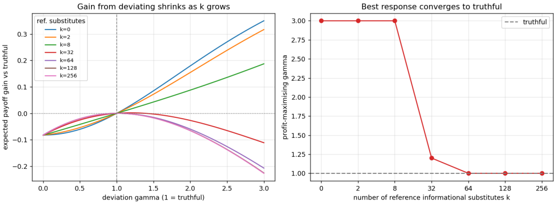
**Setup — an agent-based simulator of the mechanism**

- Binary $Y$, conditionally independent signals, payment via CE-MSR against the terminal *reference* agent.
- A focal agent deviates by scaling its signal: it reports $\mathrm{logit}(\pi) + \gamma\,\ell$, where $\ell$ is its log-likelihood ratio. $\gamma = 1$ is truthful, $\gamma > 1$ over-confident.
- The reference observes the focal report *plus* $k$ independent signals (its informational substitutes).
- Sweep $k \in \{0, 2, 8, 32, 64, 128, 256\}$ and estimate the profit-maximising $\gamma^\star$ by Monte Carlo ($4 \times 10^5$ samples).

**Hypothesis**

Few substitutes $\Rightarrow$ it pays to over-report ($\gamma^\star > 1$); as $k$ grows the reference $\approx$ ground truth, so $\gamma^\star \to 1$ (truthful).

# RQ1 Results: Truthfulness Needs an Informed Reference




- $k \leq 8$: best response is to grossly over-report ($\gamma^\star = 3$).
- $k = 32$: $\gamma^\star \approx 1.2$.
- $k \geq 64$: $\gamma^\star = 1.0$ — exactly truthful.

Matches Theorems 1–3: truthfulness emerges once the reference has enough informational substitutes.

Aside: under truthful play the market reproduces the full-information Bayesian posterior (gap $\sim 10^{-13}$).

# RQ2: What Happens When the Assumptions Break

Real Polymarket traders satisfy almost none of the paper's assumptions:

- **A1** violated — real traders are not risk-neutral, fully rational Bayesians.
- **A2** violated / unverifiable — no shared common prior or common knowledge.
- **A4** violated — price moves are serially correlated (herding, momentum, shared news), not conditionally independent given $Y$.
- **A3, A5** unverifiable / not enforced on observed price paths.

## So the theory gives *no* guarantee here

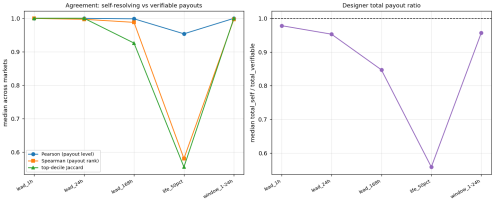
RQ2 is therefore a robustness test: does the self-resolving payout still track the verifiable one when the assumptions it relied on are broken?

# RQ2 (Empirical): Counterfactual on Real Polymarket Data

**Data & method** — $\sim$90 resolved markets (volume $>$ \$100k).

- Treat the Yes-price path as sequential reports $q^{(t)} = [1 - p_t, p_t]$; each price move is a "reporter".
- Pay the *same* transitions two ways:
  - **Self-resolving**: CE-MSR against a late reference price $r$ (no outcome needed).
  - **Verifiable**: log-MSR against the realised outcome $Y$.
- Sweep how "late" $r$ is: `lead_1h/24h/168h` (before close), `life_50pct` (halfway through the market's life), `window_1-24h` (averaged late window).
- Compare payout correlation, top-decile reporter overlap, and total payout ratio.

# RQ2 Results: Payouts Track; Leakage Drives the Gap




| ref. | Pears. | Spear. | ratio |
|---|---|---|---|
| lead_1h | 1.00 | 1.00 | 0.98 |
| lead_24h | 1.00 | 1.00 | 0.95 |
| lead_168h | 1.00 | 0.99 | 0.85 |
| life_50pct | 0.95 | 0.58 | 0.56 |
| window | 1.00 | 1.00 | 0.96 |

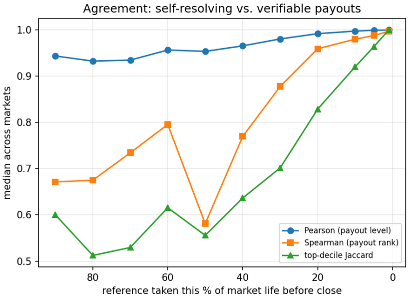
**Pearson**: do payout *amounts* agree?
**Spearman**: does the *ranking* (who earns most) agree?

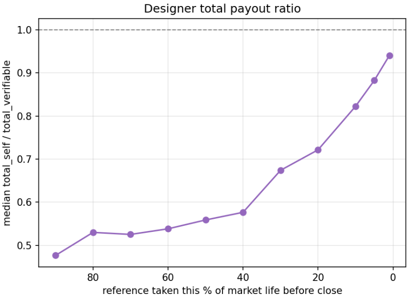
Amounts track almost perfectly; ranking degrades as $r$ moves earlier — that gap is outcome leakage.

Measures whether the *outputs* match (not incentive compatibility) — complementary to RQ1.

# RQ2: Agreement vs. *Relative* Time to Close

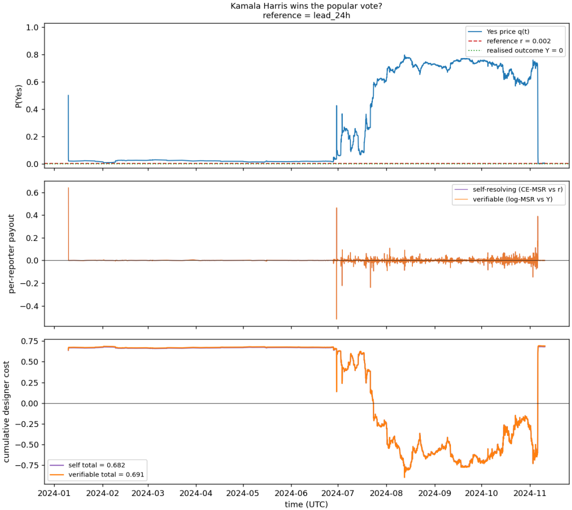
As markets can have drastically different durations, we normalize over the total duration of the market (90% before close = very early; 0% = resolution).




Agreement rises as the reference approaches close: Pearson 0.94 $\to$ 1.00, Spearman 0.67 $\to$ 1.00, payout ratio 0.48 $\to$ 0.94. Even at a genuinely early reference the payout *levels* still track (Pearson $\approx$ 0.94); the *ranking* and the designer's total cost are what the leakage channel inflates near resolution.

# RQ2: Worked Example — Kamala Harris Popular-Vote Market




A high-volume market ($\sim$2,700 price moves, outcome $Y = 0$).

The self-resolving payouts (CE-MSR vs. the 24h reference) and the verifiable payouts (vs. the outcome) are visually indistinguishable.

| | |
|---|---|
| Pearson | 1.00 |
| Spearman | 1.00 |
| payout ratio | 0.99 |
| self total | 0.68 |
| verifiable total | 0.69 |

# RQ2: Per-Market Agreement (High-Volume Markets)

Pearson / Spearman of per-reporter payouts at a *late* reference (24h before close) vs. a *mid-life* reference (halfway through the market):

| | | late (24h) | | mid-life (50%) | |
|---|---|---|---|---|---|
| Market | #rep | P | S | P | S |
| Kamala popular vote | 2681 | 1.00 | 1.00 | 1.00 | 0.45 |
| Poilievre next Canadian PM | 1030 | 1.00 | 1.00 | 0.96 | 0.91 |
| Russia–Ukraine ceasefire ’25 | 2531 | 1.00 | 1.00 | 0.99 | 0.98 |
| US government shutdown | 778 | 1.00 | 1.00 | 0.41 | −0.48 |
| TikTok ban (May ’25) | 1078 | 1.00 | 1.00 | −0.37 | −0.54 |

At a genuinely *late* reference every market tracks perfectly. At *mid-life*, some still track (ceasefire, Poilievre) while others collapse (shutdown, TikTok) — the more the late price already leaks the outcome, the larger the gap.

# Results RQ2

- Payouts track almost perfectly.
- Ranking degrades as the reference moves earlier.
- The gap is likely due to outcome leakage.
- The mechanism does seem to be able to track the correct outcome, especially at late references.

# Further questions

- Can the mechanism be adjusted so the ranking (who deserves the most reward) better matches who actually moved price toward truth, without needing the outcome?
- Can we adjust the mechanism to non binary predictions?

# The End

Questions? Comments?